HASHEM TALEB

hashem@berkeley.edu · (209) 640-9531 · San Francisco Bay Area · linkedin.com/in/hashem-h-taleb/

EDUCATION

University of California, Berkeley – *Berkeley, CA* May 2028
B.S., Electrical Engineering and Computer Sciences **GPA:** 4.0

PROFESSIONAL EXPERIENCE

Lawrence Berkeley National Laboratory – *Berkeley, CA* **Jan 2026 - Present**

Undergraduate Research Assistant (PI: Stijn Wielandt) - Part-time (Mar - Jun 2026, Aug 2026 - Present)

- Developing software for an embedded linux system and implementing radar control and signal processing algorithms.
- Build a data processing pipeline to convert raw binary radar streams into weather-radar-compatible datasets for geophysicists.
- Engineer, test, and deploy individual radar nodes to create a distributed weather radar network.
- Implemented SNR enhancement and peak detection techniques, including CFAR, to improve weak-target detection.

Electrical Engineer Intern - Full-time (Jan - Mar, Jun - Aug 2026)

- Built a deployable TI mmWave radar control platform (Raspberry Pi + Adafruit MCU, relay-switched power management, RTC based scheduling) for long-duration remote testing.
- Developed embedded/control software (C++, Bash, Python) to manage RADAR data, UART CLI, power sequencing, and watchdog recovery.
- Created a realtime Post-FFT RADAR output processing pipeline using Python and Numpy for SNR/weak-target enhancement, and reflector/beamforming validation.
- Optimized TI's mmWave DSP pipeline and implemented firmware-level reconfiguration to adapt an automotive radar for high-resolution atmospheric sensing.

Bytes.io, Co-Founder – *San Francisco, CA* **May 2021 - Apr 2025**

- Developed an AI menu generator (Python + GCP, GPT-3.5 prompting) to parse restaurant inventory and auto-create themed virtual concepts; cut onboarding time 97% and generated 1,000+ storefronts.
- Developed a central relational database to unify six fragmented data sources. Automated the ETL data processing with AWS Lambda using Python's Pandas and stored in PostgreSQL (AWS RDS), enabling analytics for 2,000+ storefronts.
- Built Google Colab tools for querying store-level metrics from a centralized database, auto-emailing reports, and distributing region-wide updates to reduce manual reporting.
- Used AWS EventBridge and Lambda to monitor critical order data across stores, accelerating issue detection and resolution.
- Built a finance reconciliation tool using Jaccard similarity to match storefront information across marketplaces, databases, and order systems, flagging low-scoring matches and cutting audit time by over 90%.
- Founded company operations by creating departments, SOPs, and systems for Sales, Marketing, and Client Onboarding.
- Designed and launched the scaling-phase service model, deploying 1,000+ storefronts across 180+ locations in CA and NY, driving Gross Merchandise Value growth to $140K/month.

PROJECTS

DC-to-AC Inverter (Power Electronics) ***Jul 2025 - Aug 2025***
Built a single-phase inverter using an H-bridge with P and N-channel MOSFETs. Built BJT gate drivers to control switching, and programmed a microcontroller to generate PWM signals emulating a sine wave at 60 Hz. Produced stable 20V, 0.5A AC output from a DC source.

Digital Logic ALU ***Jun 2024***
Designed a 4-bit Arithmetic Logic Unit (ALU) in a digital logic simulator, building from logic gates to a 4-bit adder. Implemented two's complement subtraction, carry-in for chaining adders, and status flags (zero, negative, carry).

IEC Fusion Reactor ***2026 (In-progress)***
Developing an experimental nuclear fusion device using inertial electrostatic confinement, featuring a −6 kV spherical electrode grid and grounded outer electrode within a vacuum chamber. Integrating a dual-stage vacuum pumping system to establish low-pressure plasma conditions and electrostatic ion acceleration.

SKILLS

Software Skills: Programming (Python,C/C++,Bash, SQL), Cloud Services (AWS, GCP), ERD Modeling
Hardware Skills: Raspberry Pi**,** Microcontrollers, Power electronics (Inverters, Gate drivers, Relay Switching), Communication Protocols (UART, I2C), LTspice, Altium
Language: *English: Native/Bilingual Proficiency, Arabic: Native/Bilingual Proficiency*
Interests: Embedded Systems, Wireless/RF, Digital IC, Nuclear Energy, Physics, AI/ML, Startups, Fitness & Hiking
Courses: *Comp. Architecture, Circuits, Electromagnetics, C/C++ Programming, Discrete Math & Probability, Data Structures*
HDL: Verilog (coursework in progress)